

Long-term ecological quality dynamics and spatial associations in arid and semi-arid Northwest China using an adaptive PRSEI

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Abstract

Long-term ecological monitoring in arid and semi-arid regions requires indices that capture region-specific stressors while remaining comparable through time. We developed an adaptive particular remote-sensing ecological index (PRSEI) for Gansu Province, China, using growing-season kNDVI, wetness (WET), land surface temperature (LST), sandification index (SI), and a PM10-based particulate indicator (TI) from 2000 to 2024. A unified principal component analysis (PCA) with common scaling defined the primary time series, while annual PCA and fixed weights were used for sensitivity testing. Unified PC1 explained 79.12% of total variance, and mean PRSEI increased from 0.343 to 0.419 (slope = 0.00432 yr⁻¹; R² = 0.778). Unified PCA and fixed weights agreed on 99.90% of pixel-level trend directions, whereas annual and unified PCA agreed on only 39.73%, indicating strong methodological sensitivity. Adaptive PRSEI was highly correlated with original PRSEI ($r = 0.991$) and traditional RSEI ($r = 0.975$), showing the strongest consistency with mapped restoration transitions (81.47%) but weaker consistency with degradation transitions (34.22%). Precipitation, soil type, and land use/land cover were the most stable spatial explanatory factors. Overall, ecological quality improved broadly but heterogeneously, supporting adaptive PRSEI as a complementary multi-stressor index for drylands rather than a universal replacement for RSEI.

Keywords Adaptive PRSEI · Ecological environment quality · Arid and semi-arid ecosystems · Gansu Province · Hurst exponent · Optimal-parameter geographical detector

Introduction

Ecological and environmental quality reflects multiple dimensions of ecosystem condition, including vegetation growth, surface moisture, thermal stress, land degradation, and human disturbance. Long-term ecological monitoring is particularly important in arid and semi-arid regions, where limited precipitation, high evaporative demand, sparse vegetation, and extensive bare ground increase ecosystem sensitivity to climatic variability and land-use change^{1,2}. Gansu Province, located in inland north-western China, spans the Qinghai–Tibet Plateau,

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the Inner Mongolian Plateau, and the Loess Plateau. Within a single administrative unit, it encompasses diverse landscapes, including desert–Gobi systems, irrigated oases, loess hills, alpine ecosystems, and the humid Qinba Mountains³. This pronounced environmental gradient provides a representative setting for examining spatial differences in ecological quality across contrasting dryland and mountain environments.

Remote-sensing ecological assessment has increasingly relied on composite indices that integrate multiple land-surface properties. The remote sensing ecological index (RSEI) uses principal component analysis (PCA) to combine greenness, wetness, heat, and dryness, thereby reducing reliance on subjectively assigned weights^{4–9}. Its broad application across different regions has demonstrated its effectiveness in characterising spatial variations in ecological quality. However, the conventional four-component framework may not fully capture several stressors that are particularly relevant in arid environments, including weak vegetation signals, extensive bare-ground exposure, sandification, and atmospheric particulate loading^{10–14}.

Accordingly, a growing body of research has sought to adapt ecological indices to arid environments by modifying vegetation metrics or incorporating components that better reflect dryland-specific stressors^{15–19}. In this study, the kernel normalised difference vegetation index (kNDVI) was used to capture nonlinear greenness responses, while the sandification index (SI) and a PM10-based turbidity indicator (TI) were introduced to represent additional surface and atmospheric environmental pressures. Importantly, the inclusion of additional components or a high contribution from the first principal component (PC1) does not, by itself, demonstrate superior ecological performance. A more robust evaluation therefore requires direct comparison with traditional RSEI and the original PRSEI configuration under identical years, pixels, masks, and preprocessing procedures, together with external ecological evidence to clarify where the adaptive formulation provides added information and where its limitations remain.

Long-term ecological assessments also face a fundamental challenge of temporal comparability. Analyses based on representative years provide an efficient means of comparing broad ecological differences among periods, but they may overlook interannual responses to drought, ecological restoration, irrigation, and land-cover change. By contrast, annual PCA retains year-specific information, yet the centring of variables, principal-component loadings, and score ranges may vary from year to year. Apparent temporal changes may therefore reflect not only genuine ecological variation but also changes in the index construction itself. Despite this potential source of uncertainty, few studies have directly compared annual PCA, unified PCA, and fixed-weight approaches using the same underlying dataset, leaving the sensitivity of long-term ecological trends to PCA construction insufficiently quantified.

A third methodological issue concerns the interpretation of ecological change and its explanatory factors. The Theil–Sen estimator and Mann–Kendall test are commonly used to characterise monotonic trends, the Hurst exponent describes tendencies toward persistence or reversal, and geographical detectors quantify spatial stratification^{20–24}. These methods address complementary but fundamentally different questions, and none can independently establish causal effects. In particular, the *q* statistic derived from OPGD measures the extent to which factor stratification accounts for spatial variation in ecological quality; it does not demonstrate that a given factor caused the observed changes. Robust inference therefore requires a clear distinction among descriptive patterns, temporal trends, external consistency evidence, and spatial explanatory associations.

Three research gaps emerge from the limitations outlined above. First, the extent to which a dryland-adapted PRSEI provides additional ecological information beyond the traditional RSEI has not been sufficiently evaluated under a strictly comparable framework. Second, the sensitivity of long-term PRSEI trends to different PCA construction strategies remains inadequately quantified. Third, province-wide ecological changes need to be linked to eco-regional contrasts and the robustness of spatial explanatory factors without overstating causal relationships. To address these gaps, this study constructs a temporally comparable 25-year PRSEI series using a unified PCA loading vector and common scaling, and directly compares traditional RSEI, original-configuration PRSEI, and adaptive PRSEI across identical years and common pixels. External consistency is further assessed using transition information from the China Land Cover Dataset (CLCD), while the robustness of OPGD results is

examined with respect to sampling and soil-classification schemes. Together, these analyses define the principal methodological, empirical, and interpretive contributions of this study.

Accordingly, this study aims to: (1) construct a temporally comparable adaptive PRSEI for 2000–2024; (2) quantify the sensitivity of long-term ecological trends to annual PCA, unified PCA, and fixed-weight approaches; (3) compare traditional RSEI, original-configuration PRSEI, and adaptive PRSEI in terms of spatial patterns, trend responses, and consistency with external land-cover evidence; (4) characterise ecological dynamics, class transitions, and persistence tendencies at both provincial and eco-regional scales; and (5) evaluate the stability and interaction structure of climatic, topographic, soil, land-cover, and socio-economic explanatory factors using OPGD. Throughout the analysis, CLCD transitions are treated as external categorical consistency evidence rather than as field-validated ground truth, while OPGD results are interpreted as spatial explanatory associations rather than causal mechanisms.

Materials and methods

Overview of the study area

Gansu Province is located in inland north-western China (32°11′–42°57′ N, 92°13′–108°46′ E) and covers approximately 425,800 km². The province extends across the Qinghai–Tibet Plateau, the Inner Mongolian Plateau, and the Loess Plateau, encompassing diverse landscapes such as alpine mountains, loess hills, irrigated oases, desert–Gobi systems, and mountain forests. Precipitation generally decreases from the south-east to the north-west, whereas evaporative demand and aridity increase along the same gradient. This produces a pronounced environmental transition from relatively humid mountain ecosystems in the south-east to arid oasis and desert environments in the north-west Fig. 1.

To examine regional variation in ecological quality across contrasting environmental settings, Gansu Province was divided into five eco-regions: the Hexi Desert–Gobi, Hexi Oasis–Corridor Agricultural, Longzhong–Longdong Loess Hilly, Gannan Alpine Mountain, and Longnan–Tianshui Qinba Humid Mountain regions. The regionalisation considered climatic gradients, geomorphology, land-cover patterns, ecosystem characteristics, and the spatial continuity of administrative units. These eco-region boundaries were subsequently used as consistent spatial statistical units for comparing annual PRSEI values, trend classes, and persistence tendencies Fig. 2.

Data and preprocessing

The study covered the period from 2000 to 2024. PRSEI components were derived from the MODIS/Terra MOD09A1 Version 6.1 surface reflectance product (500 m) for calculating kNDVI, WET, and SI; the MODIS/Terra MOD11A2 Version 6.1 land surface temperature product (1 km, converted from Kelvin to °C) for LST; and the Long-term Gap-free High-resolution Air Pollutants (LGHAP) v2 PM10 product (1 km, µg m⁻³) for constructing TI^{25–28}. June–September growing-season composites were used to reduce the influence of snow cover, vegetation senescence, and seasonal variation in bare-ground conditions. All continuous variables were converted to their specified physical units before spatial resampling to ensure consistency among datasets. Because the publicly available documentation for LGHAP v2 covers 2000–2021, the archived TI rasters for 2022–2024 used in this study extend beyond the documented period and are therefore explicitly acknowledged as a limitation of the data source Fig. 3.

The explanatory variables comprised annual precipitation (PRE; mm yr⁻¹), annual mean temperature (TEM; °C), potential evapotranspiration (PET; mm yr⁻¹), population density (PD; persons km⁻²), night-time lights (NTL; nW cm⁻² sr⁻¹), and annual land use/land cover (LULC), together with static layers of the digital elevation model (DEM; m), slope (SLO; °), and soil type (ST; categorical)^{29–39}. Fractional vegetation cover (FVC) was excluded from the main model because no independent annual archive was available and kNDVI was already included as

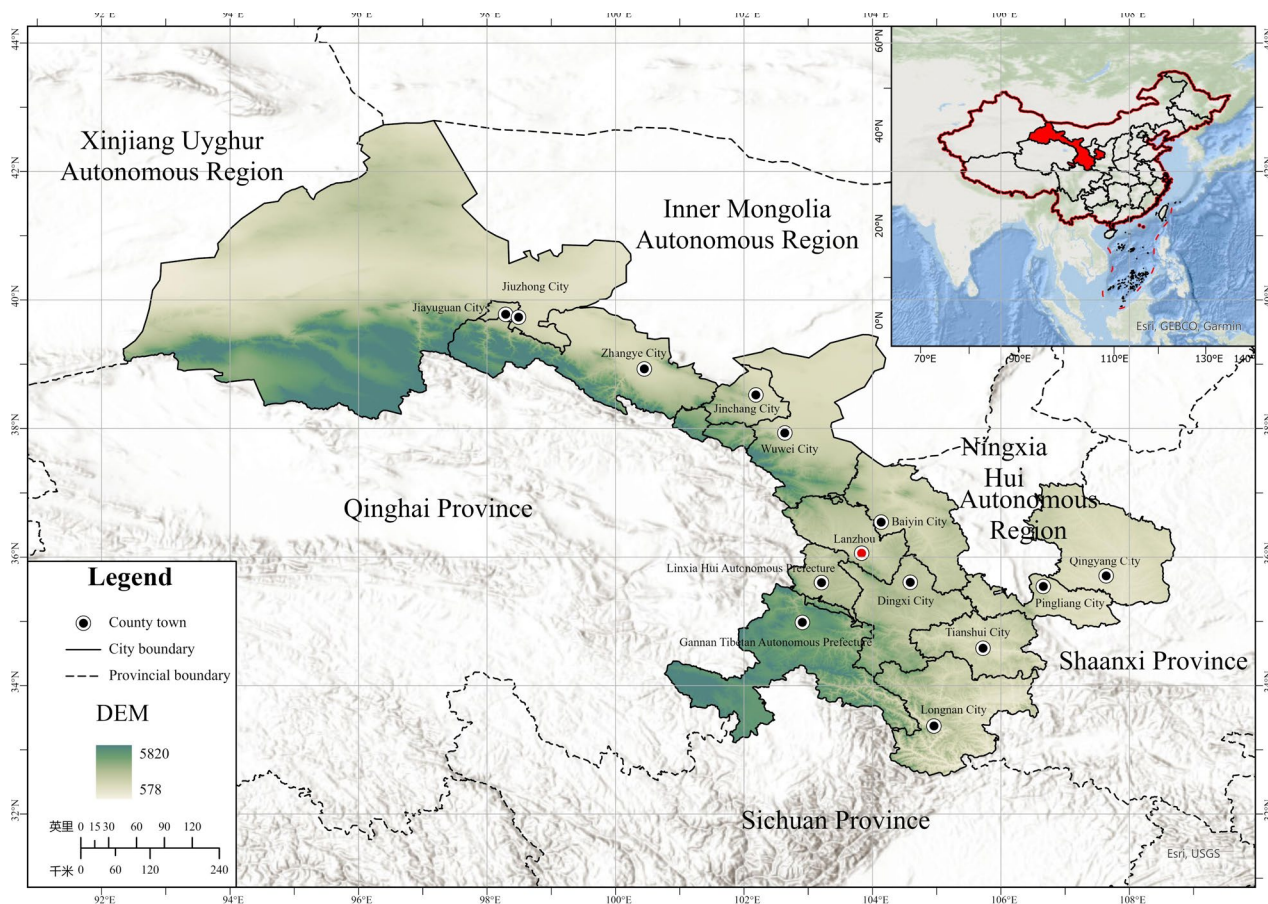


Fig. 1 Location and topographic setting of Gansu Province, China. The main map shows the provincial boundary, administrative divisions, major settlements, and elevation derived from the digital elevation model (DEM); the inset indicates the location of Gansu Province within China. Elevation is expressed in metres above sea level.

a component of PRSEI. Including both variables would introduce substantial information redundancy and could lead to circular interpretation between the response index and its explanatory factors.

All rasters were clipped to the boundary of Gansu Province, projected to the CGCS2000 3-degree Gauss–Krüger Zone 35 coordinate system (EPSG:4523), and aligned to a common grid with a spatial resolution of 474.751 m. Continuous variables were resampled using bilinear interpolation, whereas categorical variables were resampled using the nearest-neighbour method. A pixel-year observation was included in the PCA only when all required component values were finite and exceeded the NoData threshold (−9998). The primary 25-year trend and persistence analyses were conducted using a common intersection mask comprising 1,906,388 pixels. The strict three-index comparison was restricted to 24 common years (2000–2008 and 2010–2024) and 1,906,376 common pixels. The year 2009 was excluded because the data required for the three-index comparison were unavailable and was not interpolated.

Adaptive PRSEI construction

The adaptive PRSEI integrates five components: kNDVI, WET, LST, SI, and TI. Among these, kNDVI was derived from NDVI using the following nonlinear kernel transformation:

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (1)$$

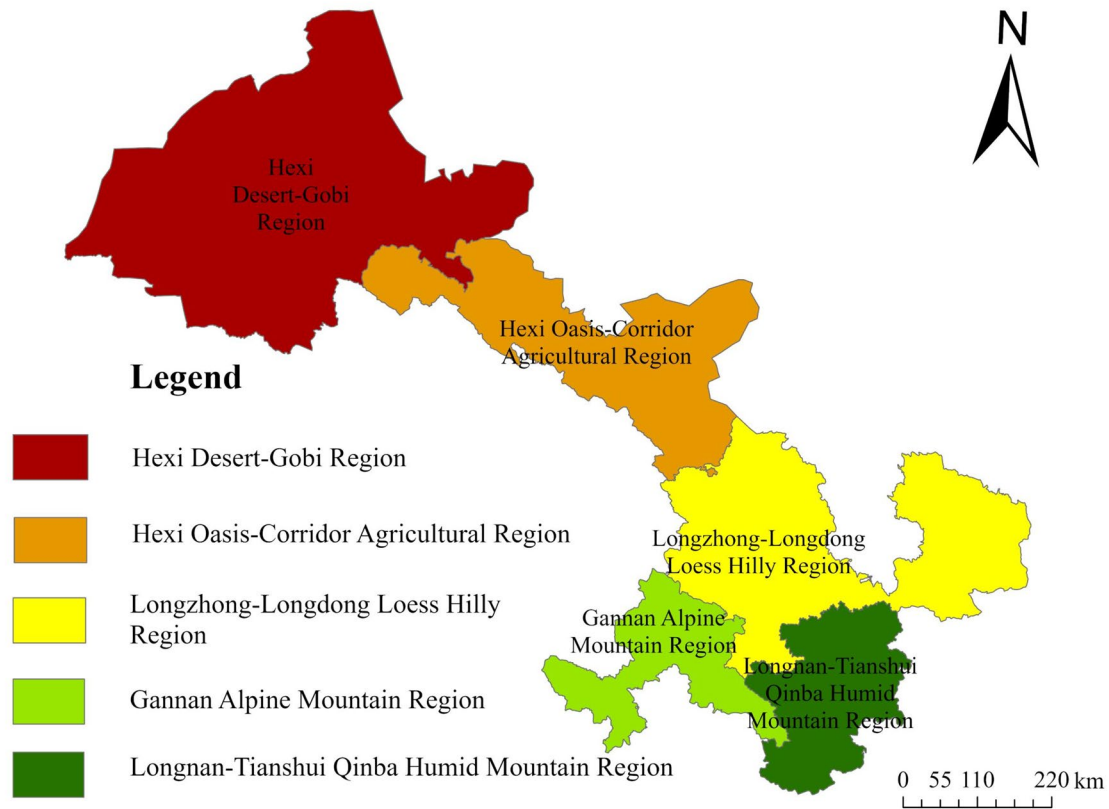


Fig. 2 Ecological regionalisation of Gansu Province used in this study. The five eco-regions are the Hexi Desert–Gobi Region, Hexi Oasis–Corridor Agricultural Region, Longzhong–Longdong Loess Hilly Region, Gannan Alpine Mountain Region, and Longnan–Tianshui Qinba Humid Mountain Region. The same boundaries were used for all eco-regional statistics.

$$kNDVI = \tanh(NDVI^2) \quad (2)$$

WET was calculated using the MODIS Tasseled Cap transformation coefficients⁴⁰:

$$WET = 0.1147B1 + 0.2489B2 + 0.2408B3 + 0.3132B4 - 0.3122B5 - 0.6416B6 - 0.5087B7 \quad (3)$$

where B1–B7 denote the corresponding MODIS surface reflectance bands. LST was derived from the MOD11A2 product using the product-specific scale factor and subsequently converted from Kelvin to degrees Celsius (°C):

$$LST(^{\circ}C) = DN \times 0.02 - 273.15 \quad (4)$$

SI was used to characterise bare-soil exposure and surface conditions associated with sandification:

$$SI = \frac{(SWIR1 + RED) - (NIR + BLUE)}{(SWIR1 + RED) + (NIR + BLUE)} \quad (5)$$

TI was derived from the mean PM10 concentration during the June–September growing season using the LGHAP v2 dataset. Because elevated particulate matter concentrations may reduce atmospheric transparency, alter surface radiative conditions, and increase particulate deposition on vegetation surfaces, higher PM10 values were assigned a negative ecological orientation in the PRSEI framework. However, PM10 should not be interpreted as a direct measure of mineral dust or sandification, as its concentration may also be influenced by combustion sources, industrial and traffic emissions, and regional atmospheric transport. Accordingly, TI is interpreted here

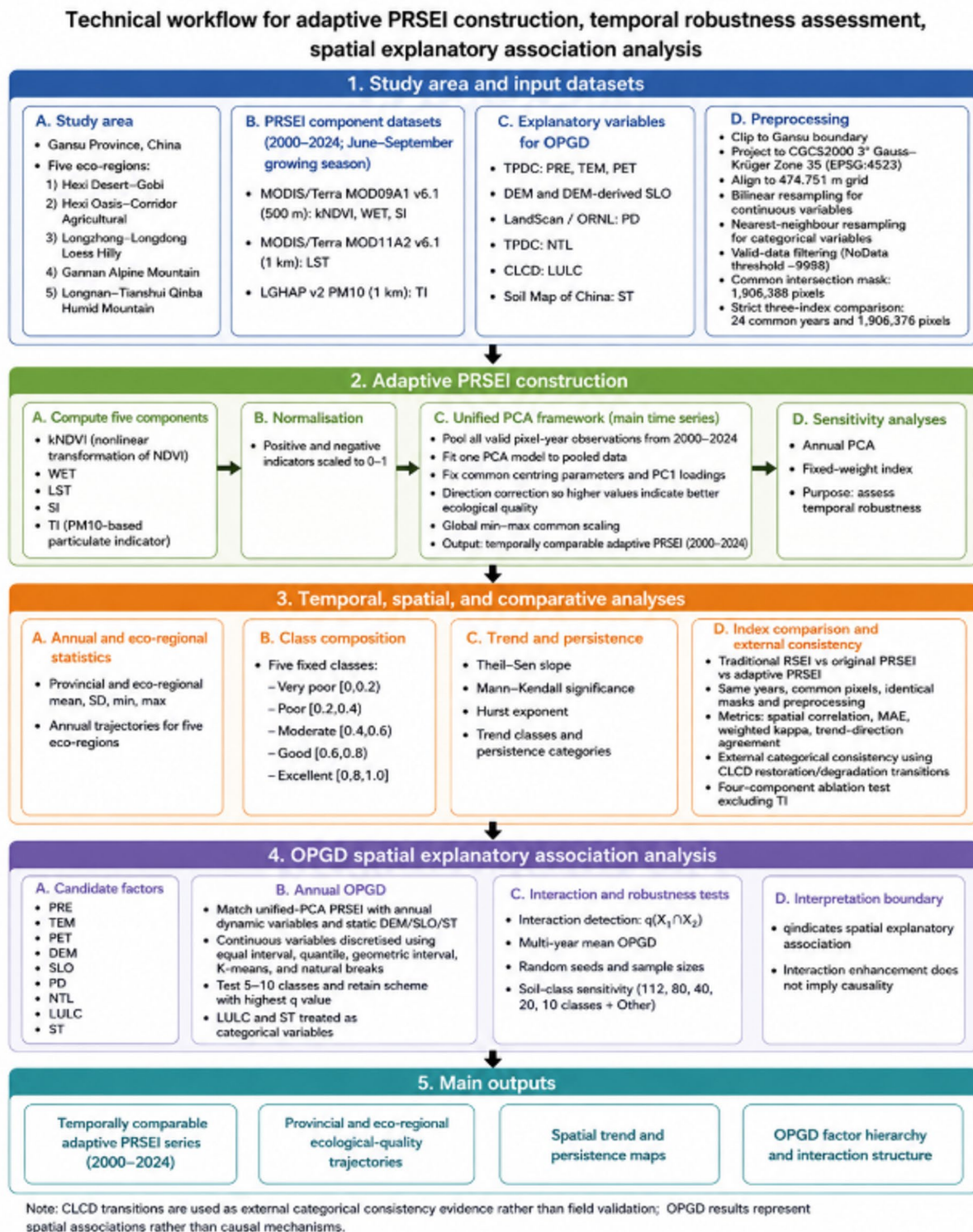


Fig. 3 Technical workflow for adaptive PRSEI construction, temporal robustness assessment, and spatial explanatory association analysis. The workflow includes data acquisition and preprocessing, unified-PCA PRSEI construction with common scaling, annual-PCA and fixed-weight sensitivity analyses, ecological-quality classification, Theil–Sen/Mann–Kendall trend analysis, Hurst-based persistence analysis, inter-index comparison, TI ablation analysis, and OPGD assessment of spatial explanatory associations.

Table 1 Datasets used for adaptive PRSEI construction and spatial explanatory association analysis.

Data group	Variable	Spatial resolution	Source	Temporal coverage	Unit
PRSEI component	kNDVI	500 m	GEE/MOD09A1 ^{25,26}	2000–2024 (Jun–Sep)	Dimensionless
PRSEI component	WET	500 m	GEE/MOD09A1 ^{25,26}	2000–2024 (Jun–Sep)	Dimensionless
PRSEI component	SI	500 m	GEE/MOD09A1 ^{25,26}	2000–2024 (Jun–Sep)	Dimensionless
PRSEI component	LST	1 km	GEE/MOD11A2 ^{25,27}	2000–2024 (Jun–Sep)	°C
PRSEI component	TI(PM10)	1 km	LGHAP ²⁸	2000–2024 (Jun–Sep)	µg m ⁻³
Explanatory factor	LULC	30 m	CLCD ²⁹	2000–2024	Categorical
Explanatory factor	ST	1:4,000,000	Soil Map of China ³⁰	Static	Categorical
Explanatory factor	PET	1 km	TPDC ³¹	2000–2024	mm yr ⁻¹
Explanatory factor	PRE	1 km	TPDC ³²	2000–2024	mm yr ⁻¹
Explanatory factor	TEM	1 km	TPDC ³³	2000–2024	°C
Explanatory factor	DEM	30 m	GEE	Static	m
Explanatory factor	SLO	30 m	DEM-derived	Static	°
Explanatory factor	NTL	500 m	TPDC ³⁴	2000–2024	nW cm ⁻² sr ⁻¹
Explanatory factor	PD	1 km	LandScan/ORNL ^{35–39}	2000–2024	persons km ⁻²

as a proxy for overall particulate matter pressure rather than a dust-specific indicator, and its association with ecological quality is not regarded as evidence of a direct causal effect.

All components were normalised to 0–1 before PCA. Positive indicators were normalised as:

$$N_i = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \text{ (positive indicators)} \quad (6)$$

Negative indicators were normalised as:

$$N_i = \frac{X_{\max} - X_i}{X_{\max} - X_{\min}} \text{ (negative indicators)} \quad (7)$$

After normalisation of the individual components, all valid pixel-year observations from 2000 to 2024 were pooled to fit a single unified PCA model. The component means and PC1 loading vector estimated from the pooled dataset were then fixed and applied consistently to each year. The orientation of PC1 was adjusted only once to ensure that higher scores consistently represented better ecological conditions. The resulting PC1 scores were subsequently rescaled using a common global minimum and maximum derived from the full 2000–2024 dataset to generate the primary PRSEI time series:

$$PRSEI = \frac{(PC1_{corr} - PC1_{corr, \min})}{(PC1_{corr, \max} - PC1_{corr, \min})} \quad (8)$$

PRSEI values range from 0 to 1, with higher values indicating ecological conditions characterised by greater greenness and wetness and lower thermal, sandification-related surface, and particulate matter pressures. Because the same centring parameters, PC1 loadings, and scaling bounds were applied consistently across all years, interannual variations in PRSEI were evaluated within a fixed index framework, thereby enhancing temporal comparability.

To assess temporal robustness, two alternative index series were additionally constructed. Annual PCA was performed independently for each year, retaining the year-specific centring parameters, PC1 loadings, and scaling procedure of the original approach. A fixed-weight index was also constructed by normalising the absolute values of the unified PC1 loadings so that the resulting weights summed to 1. The three approaches—annual PCA, unified PCA, and fixed weights—were compared in terms of annual mean values, the variance explained by PC1,

ranges of PC1 loadings, pixel-level Theil–Sen/Mann–Kendall trends, and agreement in trend direction. Because unified PCA applies a single loading vector and common scaling framework consistently across the entire study period, the unified-PCA series was adopted as the primary time series for all subsequent analyses.

The inter-index comparison used 24 common years (2000–2008 and 2010–2024) and 1,906,376 pixels shared by all datasets. Conventional RSEI comprised NDVI, WET, LST, and $1 - \text{NDBSI}$; original PRSEI comprised NDVI, WET, LST, SI, and TI; and adaptive PRSEI replaced NDVI with kNDVI. All three indices were constructed using unified PCA, common scaling, and identical spatial masks. Comparisons included provincial and eco-regional trajectories, spatial correlation, MAE, five-class agreement, weighted Cohen's κ , pixel-level trend direction, and slope correlation. CLCD restoration and degradation transitions were used only as external categorical consistency evidence, not as field validation or causal evidence.

To assess the contribution of TI, a four-component index excluding TI was constructed using the same workflow and compared with the full adaptive PRSEI in terms of spatial correlation, class agreement, provincial trend slope, and pixel-level trend-direction agreement.

Temporal, spatial and class statistics

Annual PRSEI statistics, including the mean, standard deviation, minimum, and maximum, were calculated for the province and each eco-region. Following the conventional RSEI classification and subsequent applications^{8,11,14}, PRSEI values were divided into five fixed 0.2 intervals: very poor [0, 0.2), poor [0.2, 0.4), moderate [0.4, 0.6), good [0.6, 0.8), and excellent [0.8, 1.0]. Fixed thresholds ensured direct interannual comparability and avoided artificial changes caused by year-specific quantiles or natural breaks. These classes represent relative positions along the ecological-quality gradient rather than field-calibrated ecological thresholds or regulatory standards.

Annual mean PRSEI time series for the five eco-regions were derived by overlaying the annual PRSEI rasters with the corresponding regional boundaries. These series were used to compare baseline ecological quality, the magnitude of change, and interannual variability among regions.

After applying a common valid-data mask, area statistics were derived from pixel counts using the standardised grid-cell area. For each year and eco-region, pixel counts were converted to areas and proportions for PRSEI classes, trend categories, and persistence categories. Proportional statistics were used to facilitate comparison among eco-regions of different sizes, while the underlying raster and area outputs were retained for reproducibility.

Trend and persistence analysis

Pixel-level PRSEI trends over 2000–2024 were estimated using the Theil–Sen median slope:

$$\beta = \text{Median} \left(\frac{\text{PRSEI}_j - \text{PRSEI}_i}{j - i} \right), 1 \leq i < j \leq n \quad (9)$$

Trend significance was assessed using the Mann–Kendall test, with $|Z| > 1.96$ indicating significance at $\alpha = 0.05$. Based on the Theil–Sen slope direction and Mann–Kendall significance, pixel-level trends were classified as significant decline, slight decline, stable, slight improvement, or significant improvement.

The Hurst exponent was calculated by rescaled range analysis. For each time scale n , the relationship was expressed as:

$$\log \left(\frac{R(\tau)}{S(\tau)} \right) = H \log(\tau) + C \quad (10)$$

Here, R/S denotes the rescaled range, H is the Hurst exponent, and C is a constant. Values of $H > 0.5$, $H < 0.5$, and $H = 0.5$ indicate tendencies toward persistence, reversal, and random behaviour, respectively. Combining the Hurst exponent with the Theil–Sen slope, pixels were classified as persistent improvement, improvement-to-degradation,

degradation-to-improvement, persistent degradation, or stable/uncertain. These categories represent persistence tendencies inferred from historical trajectories rather than deterministic predictions of future ecological change.

Eco-regional statistics

Using a common valid-data mask, annual PRSEI values, trend categories, and persistence tendencies were summarised for each eco-region. The main text reports key regional metrics, including initial and final values, the 25-year mean, trend slope, R^2 , and residual degradation proportion, while detailed annual statistics, class areas, and transition matrices are provided in the Supplementary Data for reproducibility without overloading the main text.

OPGD analysis of spatial explanatory associations

OPGD was used to quantify spatial associations between PRSEI and nine explanatory factors: PRE, TEM, PET, DEM, SLO, PD, NTL, LULC, and ST.

For annual OPGD analysis, unified-PCA PRSEI values were matched with the corresponding annual dynamic variables and the static DEM, SLO, and ST layers. Continuous variables were discretised using equal-interval, quantile, geometric-interval, K-means, and natural-breaks methods with 5–10 classes, and the scheme yielding the highest q value was retained. LULC and ST were treated directly as categorical variables.

The factor-detector q statistic was calculated as:

$$q = 1 - \frac{[\sum h = 1L N h \sigma h^2]}{[N \sigma^2]} \quad (11)$$

Here, h denotes the stratum, L the number of strata, Nh and N the numbers of pixels within stratum h and the full sample, and σh^2 and σ^2 the corresponding PRSEI variances. A higher q value indicates a greater proportion of PRSEI spatial variance explained by factor stratification. Accordingly, q is interpreted here as a measure of spatial explanatory association rather than a regression coefficient, causal effect, or test of causality.

Interaction detection was performed by overlaying pairs of factor strata and comparing $q(X1 \cap X2)$ with the corresponding single-factor q values. A higher interaction q value indicates stronger joint explanatory association than either factor alone, but does not identify the underlying mechanism or quantify the ecological effect of changes in individual factors.

For reproducibility, the selected discretisation method and class number for each continuous variable were recorded, while LULC and ST retained their original categories. To assess sensitivity to soil classification, the original 112 ST classes were progressively consolidated into the 80, 40, 20, and 10 most frequent classes, with the remaining classes grouped as “Other”, and q values were recalculated for each scheme.

Robustness of the multi-year OPGD results was further evaluated across random seeds (42, 512, and 2024), sample sizes (25,000, 50,000, and 100,000), four discretisation methods, and 5–10 classes. Stability was assessed by comparing factor rankings, overlap among the top three factors, and q values of key interactions across settings. These tests quantify sensitivity to sampling and discretisation choices rather than causal effects.

Results

Stability of annual PRSEI construction

Unified PCA defined a temporally consistent composite axis using pooled observations from 2000 to 2024. PC1 explained 79.12% of the total variance, with direction-corrected loadings of 0.709 for kNDVI, 0.138 for WET, 0.388 for LST, 0.342 for SI, and 0.459 for TI (Table 2). These loadings quantify the contributions of the

standardised components to the composite axis but do not, by themselves, demonstrate the ecological superiority of a particular index formulation.

Annual PCA showed substantial temporal variation in index structure. PC1 explained 78.37–85.70% of the variance (mean 81.82%), while loadings ranged from 0.628 to 0.770 for kNDVI, 0.132–0.154 for WET, 0.338–0.442 for LST, 0.293–0.386 for SI, and 0.326–0.554 for TI. This variability indicates that trends derived from annual PCA may reflect both ecological change and year-to-year shifts in component weighting.

Sensitivity analysis revealed strong dependence of long-term trend interpretation on PCA construction. Annual PCA with year-specific scaling classified 39.21% of pixels as improving, 3.34% as stable, and 57.45% as degrading, compared with 99.42%, 0.07%, and 0.52% under unified PCA and 99.36%, 0.07%, and 0.57% under the fixed-weight approach, respectively. Pixel-level trend-direction agreement reached 99.90% between unified PCA and fixed weights but only 39.73% between annual and unified PCA. Thus, the apparent widespread degradation inferred from annual PCA was strongly method-sensitive, and subsequent analyses were based on the unified-PCA series.

Provincial annual dynamics and ecological-quality classes

Based on the unified-PCA series, the mean PRSEI for Gansu Province increased from 0.343 in 2000 to 0.419 in 2024, with a linear slope of 0.00432 yr⁻¹ (R² = 0.778), indicating a clear long-term improvement despite interannual variability. Figure 4 presents the provincial mean PRSEI derived from unified PCA together with the corresponding ecological-quality class composition. Annual-PCA results are used only for sensitivity assessment and are not used for primary temporal inference.

Class transitions based on fixed thresholds provide a more conservative measure of structural change than trend direction alone. From 2000 to 2024, 37.57% of pixels increased by at least one 0.2 class, 60.73% remained unchanged, and 1.71% decreased, indicating that widespread positive trends did not necessarily translate into shifts across ecological-quality classes. Provincial trends should therefore be interpreted alongside baseline conditions and class transitions. Common scaling ensured temporally consistent thresholds, while the five classes were treated as relative mapping intervals rather than field-calibrated ecological-status thresholds.

Spatial pattern and eco-regional divergence

Unified PCA revealed a clear south-east–north-west gradient in ecological quality. The 25-year mean PRSEI was highest in the Longnan–Tianshui Qinba Humid Mountain Region (0.692), followed by the Gannan Alpine Mountain Region (0.672), Longzhong–Longdong Loess Hilly Region (0.440), Hexi Oasis–Corridor Agricultural Region (0.343), and Hexi Desert–Gobi Region (0.225). This spatial ranking remained consistent across representative years, confirming a stable pattern of higher ecological quality in south-eastern Gansu and lower values toward the arid north-west Fig. 5.

All five eco-regions showed positive PRSEI trends, but improvement rates varied markedly. Slopes were highest in the Longnan–Tianshui Qinba Humid Mountain Region (0.00764 yr⁻¹), followed by the Longzhong–Longdong Loess Hilly Region (0.00649 yr⁻¹), Gannan Alpine Mountain Region (0.00481 yr⁻¹), Hexi Oasis–Corridor Agricultural Region (0.00310 yr⁻¹), and Hexi Desert–Gobi Region (0.00270 yr⁻¹). Thus, the Hexi Desert–Gobi

Table 2 Unified-PCA loadings and interannual variability in annual-PCA statistics for the adaptive PRSEI during 2000–2024.

Indicator	Minimum	Maximum	Mean	Unified PCA
PC1 contribution (%)	78.37	85.70	81.82	79.12%
kNDVI loading	0.628	0.770	0.717	0.709
WET loading	0.132	0.154	0.141	0.138
LST loading	0.338	0.442	0.399	0.388
SI loading	0.293	0.386	0.346	0.342
TI loading	0.326	0.554	0.427	0.459

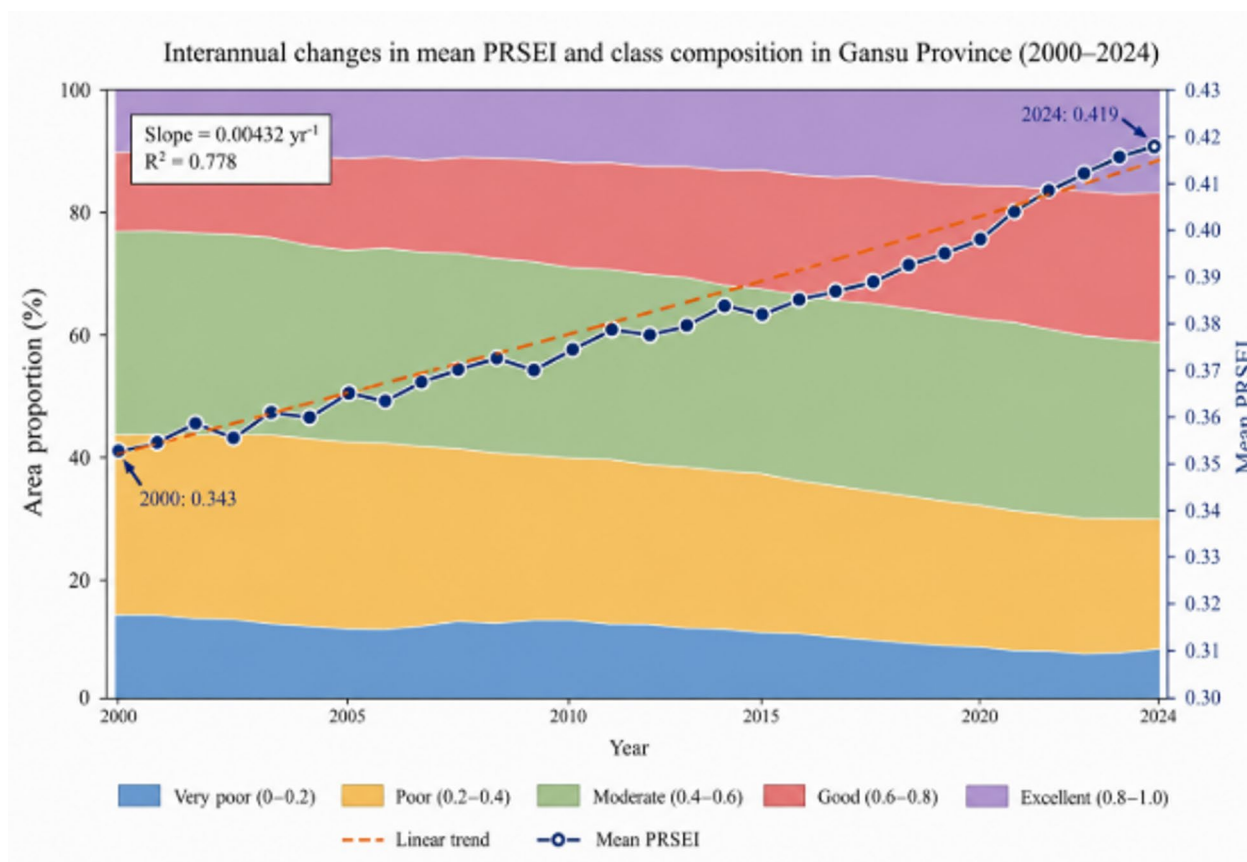


Fig. 4 Annual mean unified-PCA PRSEI and ecological-quality class composition in Gansu Province from 2000 to 2024. The line represents the annual provincial mean PRSEI, and the dashed line denotes its linear trend (slope = 0.00432 yr⁻¹; R² = 0.778). Stacked areas show the proportions of five fixed PRSEI classes: very poor [0, 0.2), poor [0.2, 0.4), moderate [0.4, 0.6), good [0.6, 0.8), and excellent [0.8, 1.0]. PRSEI is dimensionless and class proportions are expressed as percentages.

Region combined the lowest ecological-quality baseline with the slowest improvement, while the Hexi Oasis–Corridor Agricultural Region retained the largest share of negative trends (2.07%). These results indicate that broad ecological improvement has not eliminated pronounced regional disparities Fig. 6.

Historical trend classes

Under the temporally comparable unified-PCA framework, 99.42% of the common-mask area showed positive trends, 0.07% remained stable, and only 0.52% exhibited negative trends. This pattern contrasts sharply with the annual-PCA results, confirming that the earlier inference of widespread degradation was strongly sensitive to PCA construction Fig. 7.

Positive trends were widespread across all five eco-regions, although their ecological significance varied with baseline conditions and class transitions. Residual negative trends were highest in the Hexi Oasis–Corridor Agricultural Region (2.07%), followed by the Longnan–Tianshui Qinba Region (0.33%), and remained below 0.20% elsewhere. Because 60.73% of pixels remained within the same 0.2 PRSEI class, the overall provincial pattern is better characterised as widespread but heterogeneous improvement rather than universal recovery. Accordingly, subsequent interpretation considers trend direction, slope magnitude, class transitions, and baseline PRSEI jointly to avoid overstatement.

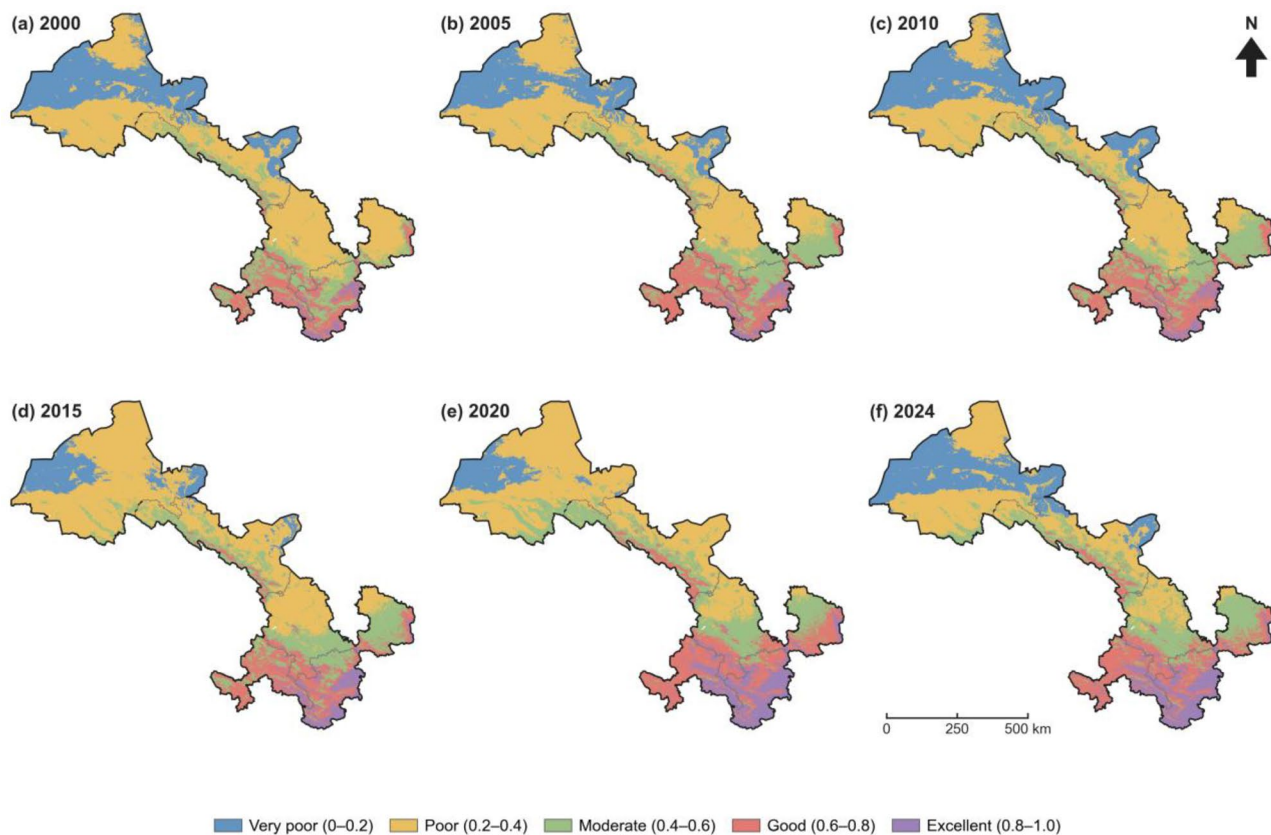


Fig. 5 Spatial distribution of unified-PCA PRSEI ecological-quality classes in representative years across Gansu Province. Maps are shown for (a) 2000, (b) 2005, (c) 2010, (d) 2015, (e) 2020, and (f) 2024. All years use common scaling and the same fixed class thresholds: very poor [0, 0.2), poor [0.2, 0.4), moderate [0.4, 0.6), good [0.6, 0.8), and excellent [0.8, 1.0].

Persistence tendencies from Hurst coupling

Coupling unified-PCA trends with the Hurst exponent showed that 99.415% of the area exhibited persistent improvement, 0.517% persistent degradation, and 0.066% stable or uncertain behaviour, while reversal categories together accounted for <0.002%. These results indicate persistence tendencies in historical trajectories rather than deterministic predictions of future ecological change Fig. 8.

Residual degradation was highest in the Hexi Oasis–Corridor Agricultural Region (2.062%), followed by the Longnan–Tianshui Qinba Humid Mountain Region (0.328%), Longzhong–Longdong Loess Hilly Region (0.187%), Gannan Alpine Mountain Region (0.139%), and Hexi Desert–Gobi Region (0.127%). Although these proportions were small, their occurrence across eco-regions with contrasting baseline conditions warrants targeted monitoring. Accordingly, the Hurst analysis is used here as a monitoring indicator of historical persistence rather than a predictive tool; a positive trend combined with $H > 0.5$ indicates persistence in past trajectories but does not guarantee continued improvement under future climatic, water-use, or land-cover change.

OPGD single-factor results

In the annual unified-PCA OPGD analysis, PRE showed the highest mean q value (0.829), followed by ST (0.790), LULC (0.677), PET (0.531), SLO (0.443), PD (0.299), DEM (0.149), TEM (0.115), and NTL (0.007) (Table 3). PRE, ST, and LULC consistently ranked among the top three factors across all annual analyses. These q values represent spatial explanatory associations rather than causal effects.

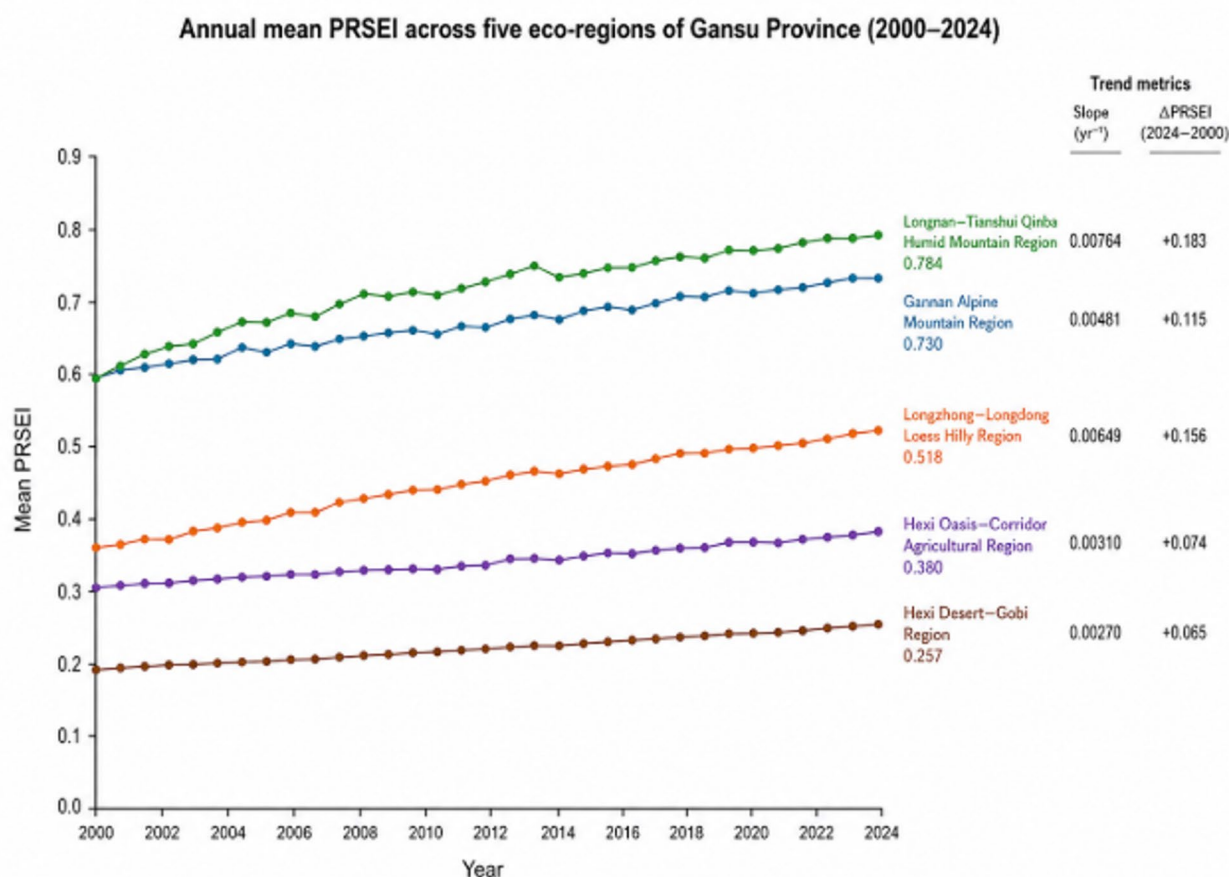


Fig. 6 Annual mean unified-PCA PRSEI trajectories for the five eco-regions of Gansu Province from 2000 to 2024. The five series represent the Hexi Desert–Gobi, Hexi Oasis–Corridor Agricultural, Longzhong–Longdong Loess Hilly, Gannan Alpine Mountain, and Longnan–Tianshui Qinba Humid Mountain regions. Labels at the right indicate the 2024 PRSEI values, linear trend slopes (yr⁻¹), and net PRSEI changes between 2000 and 2024 (ΔPRSEI). PRSEI is dimensionless.

The stable factor ranking supports a precipitation–soil–land-cover framework for spatial variation in PRSEI: precipitation covaries with vegetation and surface moisture, soil type differentiates water-retention and erosion conditions, and LULC captures major surface-background differences. These processes provide plausible ecological context for the observed spatial stratification, although OPGD cannot identify temporal pathways or causal intervention effects. Robustness tests showed that the leading factors remained stable across random seeds, sample sizes, and model settings. For ST, q decreased from 0.800 with all 112 classes to 0.795, 0.716, 0.592, and 0.513 after consolidation to the top 80, 40, 20, and 10 classes, respectively, indicating that soil remains an important spatial explanatory factor but that its q magnitude is sensitive to classification granularity. The multi-year OPGD analysis yielded the same leading factors (PRE=0.854, ST=0.801, LULC=0.688; Fig. 9), further supporting the robustness of the overall ranking.

OPGD interaction results

Two-factor interactions consistently showed higher q values than the corresponding single factors. Among annual mean interactions, PRE∩ST was strongest ($q=0.907$), followed by LULC∩ST ($q=0.898$) and PRE∩LULC ($q=0.889$), indicating a stable precipitation–soil–land-cover interaction structure in PRSEI spatial heterogeneity. The multi-year interaction matrix showed the same dominant combinations, with PRE∩ST ($q=0.921$),

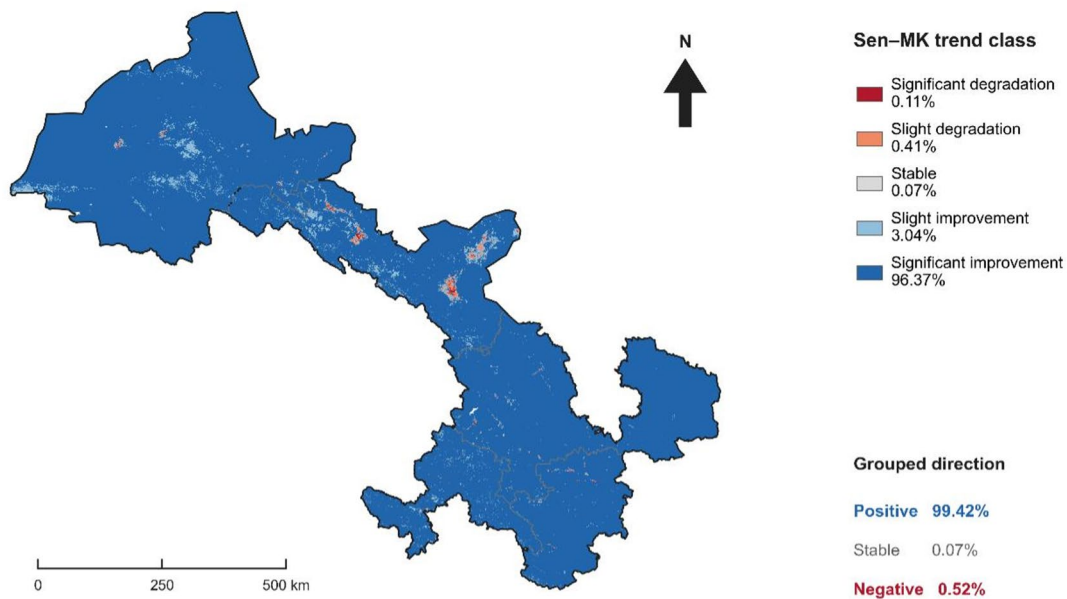


Fig. 7 Spatial distribution of Theil–Sen/Mann–Kendall PRSEI trend classes in Gansu Province from 2000 to 2024. Trends were derived from the unified-PCA PRSEI series and classified as significant decline, slight decline, stable, slight improvement, or significant improvement according to Theil–Sen slope direction and Mann–Kendall significance at $\alpha=0.05$. Positive, stable, and negative trends account for 99.42%, 0.07%, and 0.52% of the common-mask area, respectively.

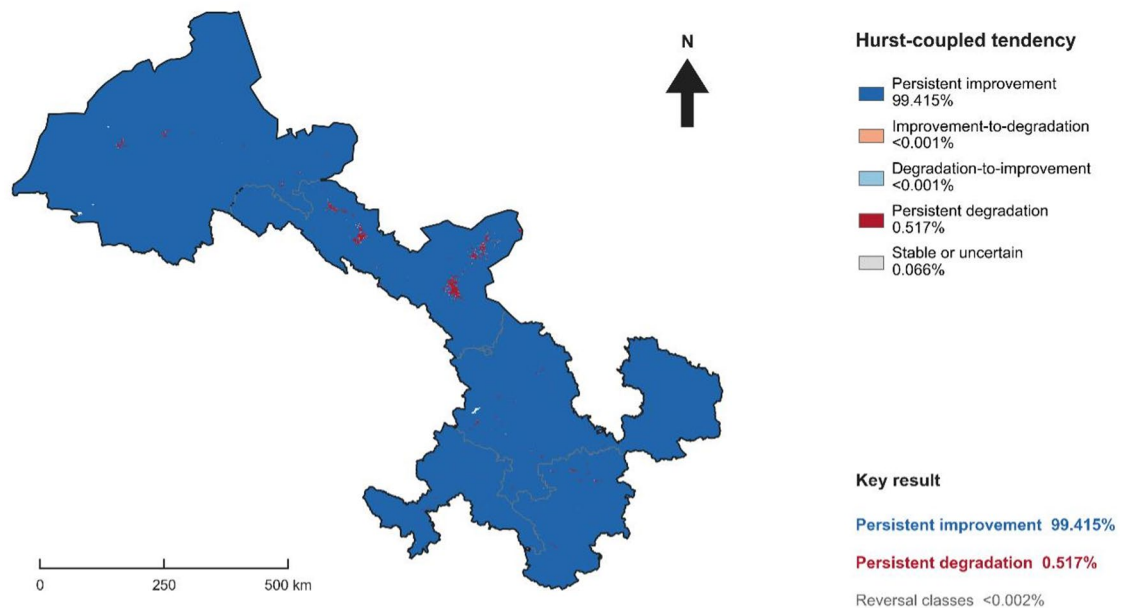


Fig. 8 Spatial distribution of Hurst-coupled PRSEI persistence tendencies in Gansu Province. Unified-PCA Theil–Sen trends were coupled with the Hurst exponent to classify pixels as persistent improvement, improvement-to-degradation, degradation-to-improvement, persistent degradation, or stable/uncertain. These classes describe persistence tendencies in historical trajectories and should not be interpreted as deterministic predictions of future ecological change.

Table 3 Single-factor OPGD results for the multi-year mean PRSEI.

Rank	Factor	q value	Variable type	Optimal method	Classes
1	PRE	0.854	Continuous	Quantile	10
2	ST	0.801	Categorical	Original categories	112
3	LULC	0.688	Categorical	Original categories	10
4	PET	0.541	Continuous	Quantile	10
5	SLO	0.453	Continuous	K-means	10
6	PD	0.319	Continuous	Quantile	10
7	DEM	0.149	Continuous	K-means	10
8	TEM	0.120	Continuous	K-means	10
9	NTL	0.019	Continuous	Geometric interval	5

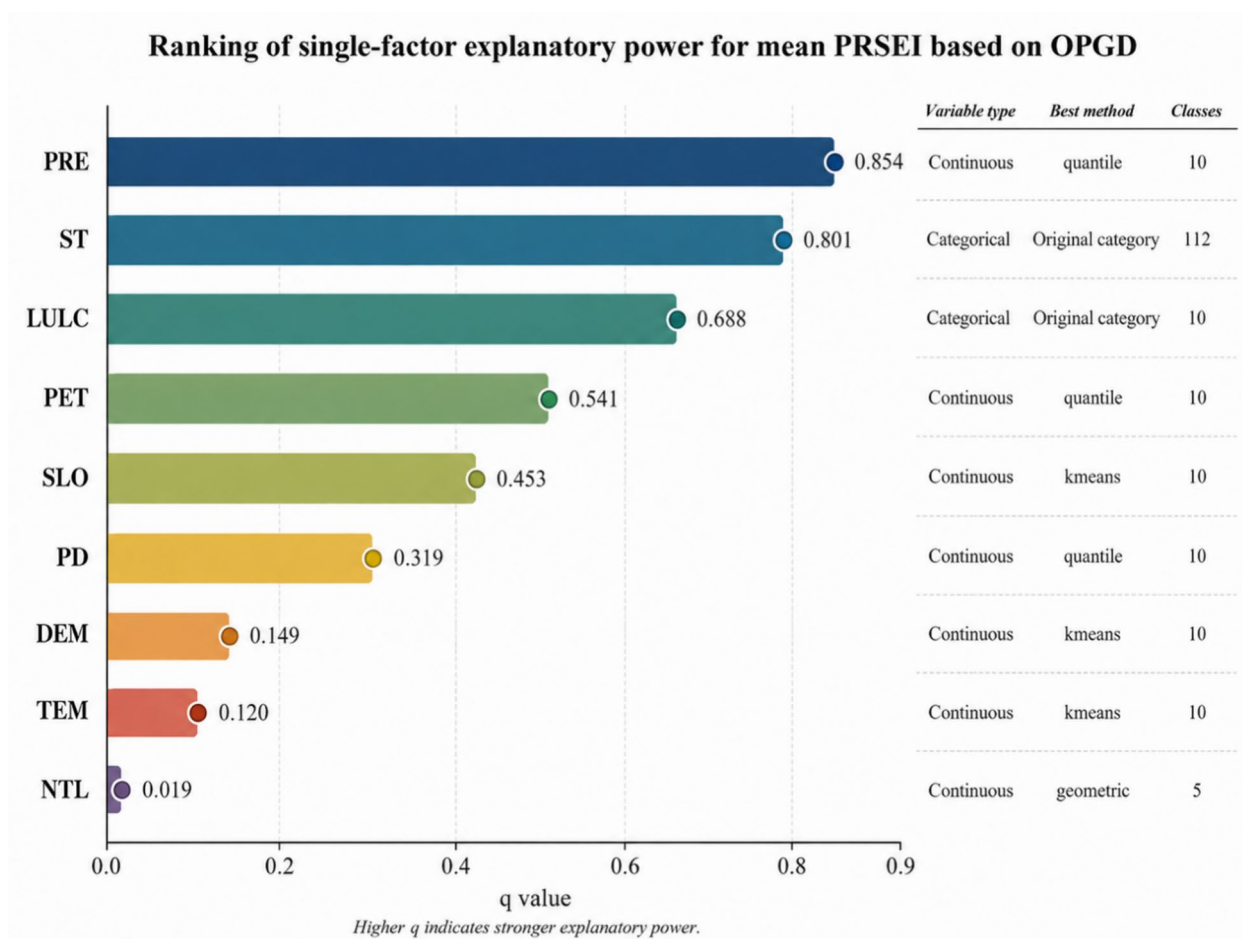


Fig. 9 Single-factor OPGD explanatory power for the multi-year mean PRSEI. Bars show factor-detector q values for PRE, ST, LULC, PET, SLO, PD, DEM, TEM, and NTL. The accompanying information reports variable type, optimal discretisation method, and number of classes. The q statistic is dimensionless and represents spatial explanatory association rather than causal effect.

PRE $\cap$ LULC ($q=0.909$), and ST $\cap$ LULC ($q=0.908$) ranking highest (Figs. 10 and 11). These interaction enhancements indicate stronger joint spatial explanatory associations, but do not imply causal relationships or quantify the ecological effects of changes in individual factors.

Annual q-value trajectories showed a clear hierarchy of spatial associations, with PRE, ST, and LULC forming the leading tier, PET and SLO the second tier, and PD, DEM, TEM, and NTL showing weaker independent associations. The very low q value of NTL should not be interpreted as evidence of negligible human influence,

because irrigation, grazing, groundwater abstraction, and ecological restoration are poorly captured by night-time lights alone. Although factor rankings were generally stable, q magnitudes remained sensitive to spatial resolution, discretisation, and variable selection. OPGD results are therefore interpreted as patterns of spatial association and interaction enhancement rather than causal mechanisms. Overall, water availability, soil background, and land-cover configuration emerge as priorities for future process-based validation, while the current evidence is insufficient to attribute PRSEI changes to any single factor or specific policy intervention Fig. 12.

Comparison with traditional RSEI and original-configuration PRSEI

Across the 24 common years, adaptive PRSEI showed very high spatial consistency with original PRSEI (mean $r=0.991$, $MAE=0.045$, five-class agreement=75.61%, weighted $\kappa=0.911$). Its correlation with traditional RSEI was also strong ($r=0.975$), although class agreement was lower (60.20%; weighted $\kappa=0.868$). Traditional RSEI and original PRSEI were similarly correlated ($r=0.986$), with 69.43% class agreement. Overall, the three indices captured broadly similar spatial gradients but differed appreciably in ecological-quality classification.

Provincial trend slopes were 0.00141 yr^{-1} for RSEI, 0.00340 yr^{-1} for original PRSEI, and 0.00429 yr^{-1} for adaptive PRSEI. Trend-direction agreement was 99.00% between adaptive and original PRSEI ($r=0.932$ for pixel-level slopes), but only 51.15% between adaptive PRSEI and RSEI ($r=0.895$). At the eco-regional scale, traditional RSEI showed slight negative trends in the Hexi Desert–Gobi and Gannan Alpine Mountain regions,

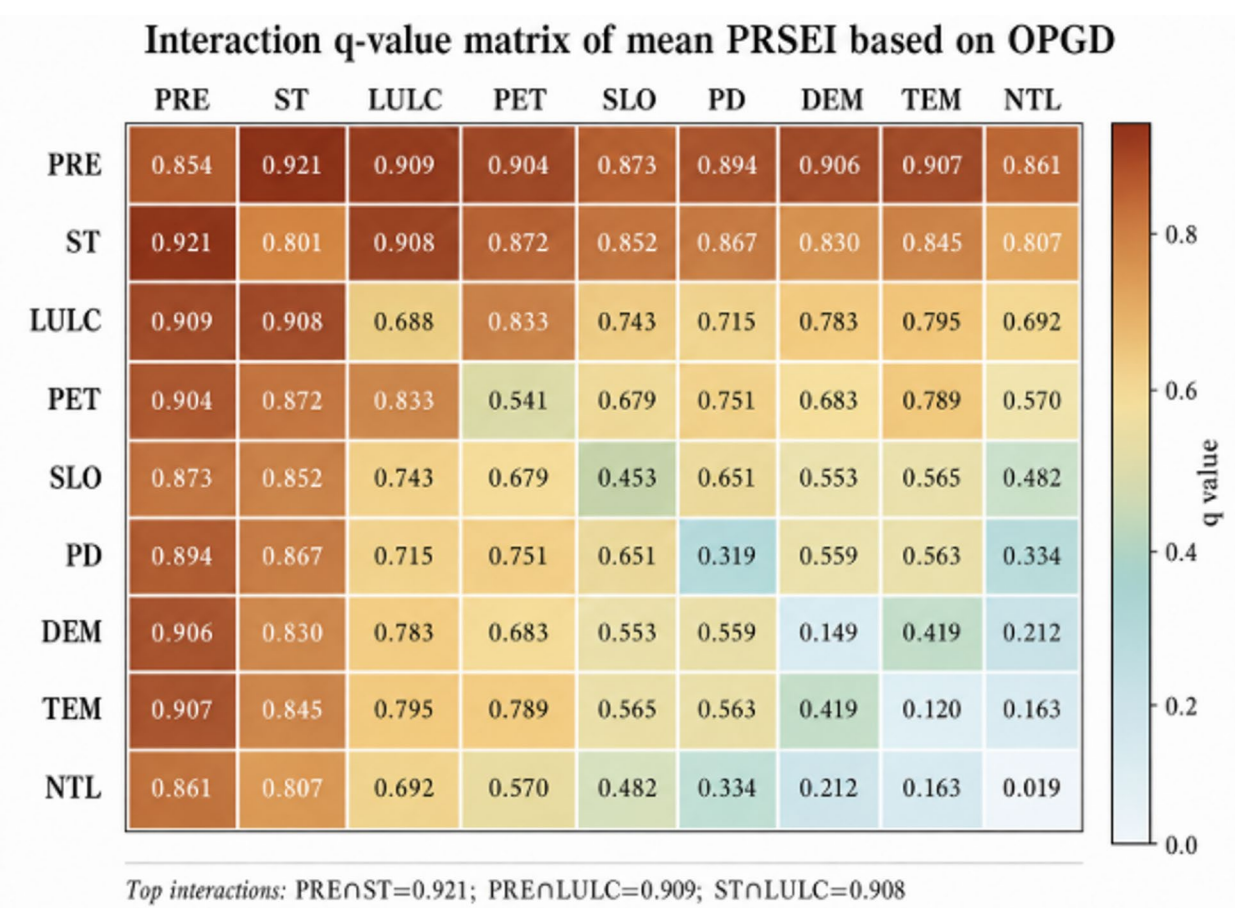


Fig. 10 Pairwise OPGD interaction q -value matrix for the multi-year mean PRSEI. Diagonal cells show single-factor q values, whereas off-diagonal cells show interaction values $q(X_1 \cap X_2)$ for pairs of explanatory factors. Higher q values indicate stronger spatial explanatory associations. The q statistic is dimensionless and does not represent a causal effect.

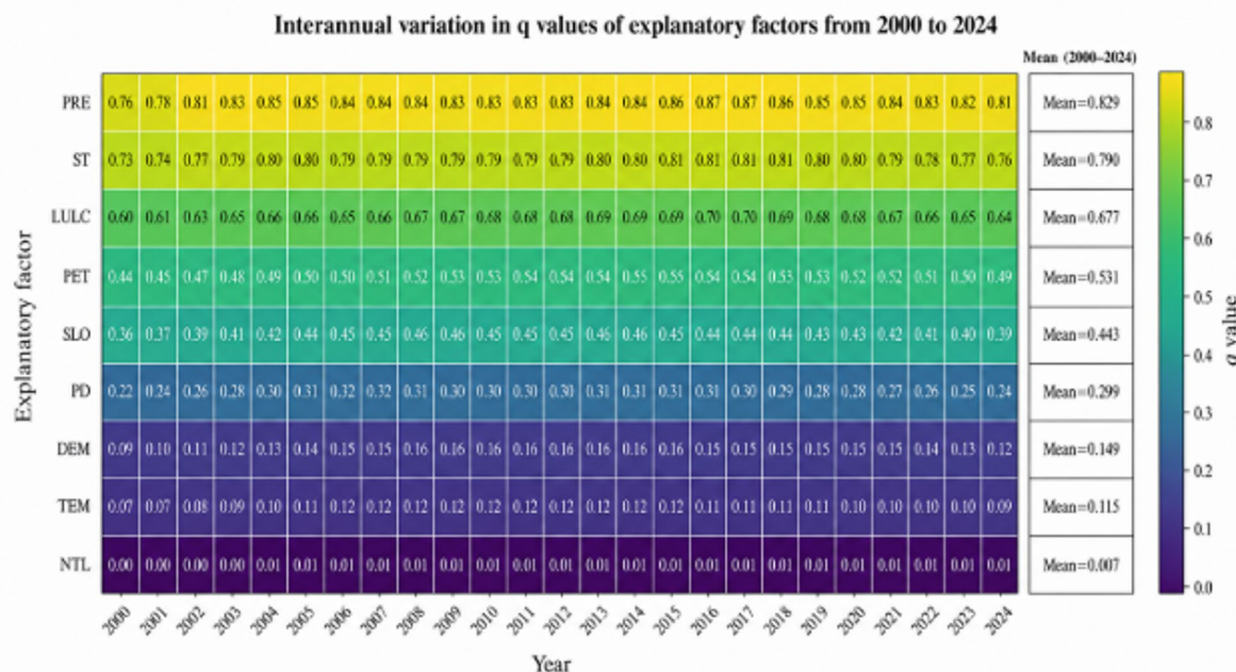


Fig. 11 Interannual variation in OPGD q values of explanatory factors in Gansu Province from 2000 to 2024. Each cell shows the annual factor-detector q value, and the right-hand column reports the corresponding multi-year mean. Factors are ordered by their mean q values. The q statistic is dimensionless and represents the proportion of spatial PRSEI variance associated with factor stratification.

whereas both PRSEI formulations showed positive trends across all five eco-regions. These contrasts indicate that long-term trend inference is sensitive to index composition Fig. 13.

External land-cover consistency revealed a clear trade-off among the three indices. In CLCD restoration-transition areas, positive changes were identified in 55.58% of pixels by RSEI, 78.97% by original PRSEI, and 81.47% by adaptive PRSEI. In degradation-transition areas, negative changes occurred in 70.36%, 44.53%, and 34.22% of pixels, respectively. Thus, adaptive PRSEI showed the highest consistency with mapped restoration but the lowest with degradation. Overall balanced consistency was 62.97% for RSEI, 61.75% for original PRSEI, and 57.84% for adaptive PRSEI, indicating that no index was consistently superior across both transition types Fig. 14.

All three indices clearly distinguished stable natural vegetation from stable bare ground. Cohen's d values for 2000 and 2024 were 2.019 and 2.408 for RSEI, 2.227 and 2.800 for original PRSEI, and 1.923 and 2.454 for adaptive PRSEI, respectively. However, adaptive PRSEI showed the smallest separation from original PRSEI within the lowest-NDVI group, providing no evidence that it improves discrimination of sparse vegetation.

Discussion

Comparative performance and trade-offs of adaptive PRSEI

Direct comparison among the three indices provides a more robust basis for evaluating adaptive PRSEI than PC1 variance contribution alone. Adaptive PRSEI remained highly consistent with original PRSEI in both spatial pattern ($r=0.991$) and trend direction (99.00% agreement), indicating that replacing NDVI with kNDVI modified

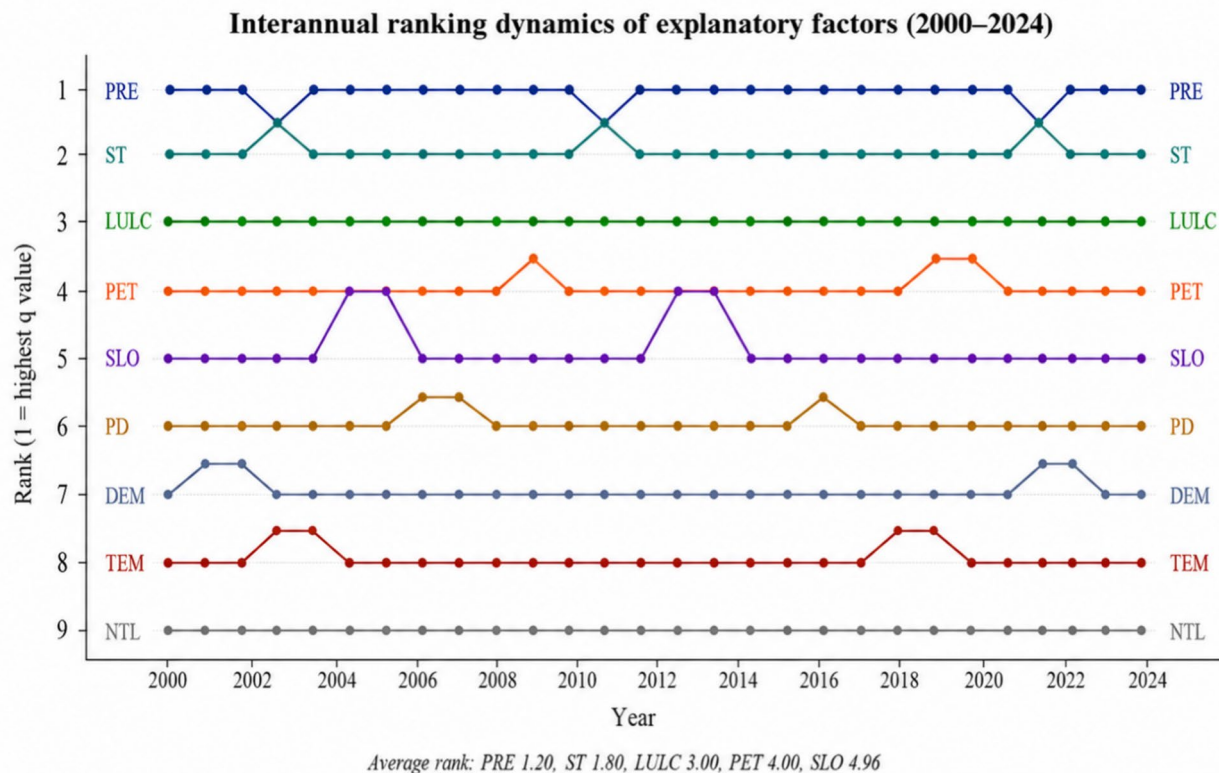


Fig. 12 Interannual ranking dynamics of OPGD explanatory factors in Gansu Province from 2000 to 2024. Annual ranks were determined from factor-detector q values, with rank 1 representing the highest q value. Labels at the bottom indicate the mean ranks over the study period. Rankings describe relative spatial explanatory associations rather than causal importance.

component balance without fundamentally altering the index structure. Its spatial correlation with traditional RSEI was also high ($r=0.975$), but trend-direction agreement was much lower (51.15%), highlighting the sensitivity of long-term change trajectories to index composition.

CLCD-based external consistency further revealed a trade-off rather than universal superiority. Adaptive PRSEI showed the highest consistency with restoration transitions (81.47%) but the lowest with degradation transitions (34.22%), compared with 78.97% and 44.53% for original PRSEI and 55.58% and 70.36% for RSEI, respectively. All three indices also distinguished stable natural vegetation from stable bare land. These results support adaptive PRSEI as a complementary multi-stressor index for drylands, particularly for capturing recovery-related responses, while degradation assessment should be corroborated using RSEI, land-cover transitions, or independent field evidence.

Two component-level findings further refine interpretation of the adaptive PRSEI. First, the difference between adaptive and original PRSEI was not greatest in the lowest-NDVI group, providing no evidence that kNDVI improves sparse-vegetation discrimination; it is therefore interpreted here only as a nonlinear greenness transformation. Second, excluding TI reduced the provincial trend slope from 0.00432 to 0.00248 yr^{-1} , while spatial correlation remained high ($r=0.988$) and trend-direction agreement was 75.97%, indicating that TI materially affects temporal trend inference. However, PM10 is treated as a general particulate-matter proxy rather than a dust-specific indicator, and dust-related mechanisms require further validation using independent evidence such as AOD, dust-event frequency, or deposition data.

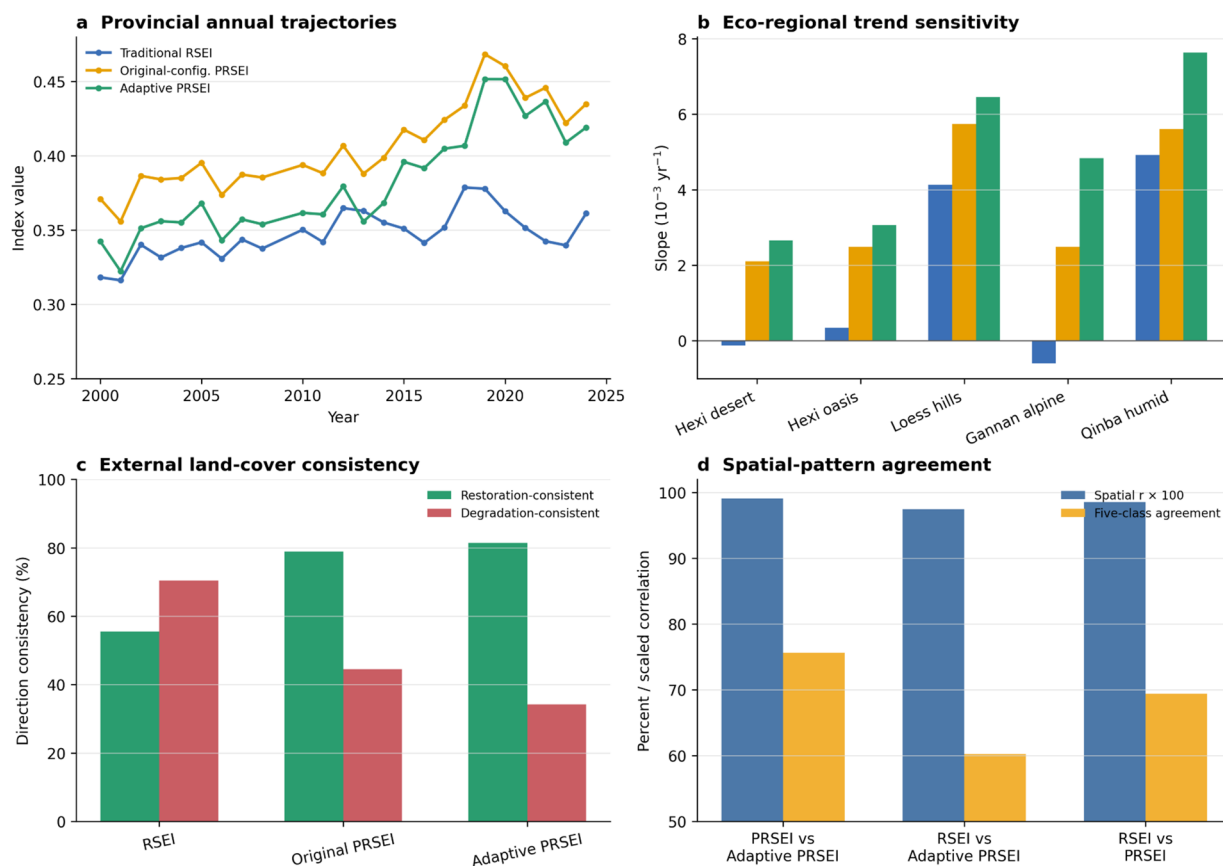


Fig. 13 Comparison of traditional RSEI, original-configuration PRSEI, and adaptive PRSEI over the 24 common years (2000–2008 and 2010–2024). (a) Provincial annual trajectories; (b) eco-regional linear trend slopes; (c) consistency of index-change direction with CLCD restoration and degradation transitions; and (d) pairwise spatial correlation and five-class agreement. All comparisons use identical years and valid-pixel masks. Index values, correlation coefficients, and agreement rates are dimensionless; trend slopes are expressed in yr^{-1} .

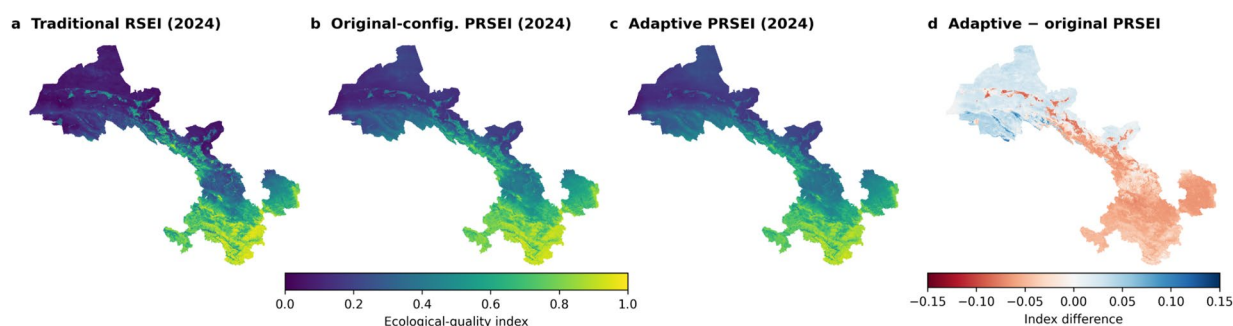


Fig. 14 Spatial comparison of the three ecological indices in 2024. (a) Traditional RSEI, (b) original-configuration PRSEI, (c) adaptive PRSEI, and (d) the difference between adaptive and original PRSEI. All indices were constructed using unified PCA fitted to pooled observations, common scaling, and the same valid-pixel mask. Index values and differences are dimensionless.

Temporal comparability reshapes long-term trend interpretation

The contrast between annual and unified PCA demonstrates that long-term ecological trend inference is highly sensitive to index construction. Because annual PCA allows component loadings, centring parameters, and scaling ranges to vary among years, apparent temporal changes may partly reflect changes in the composite index itself. The high trend-direction agreement between unified PCA and fixed weights (99.90%), compared with only 39.73% between annual and unified PCA, supports the use of a fixed index framework for long-term temporal inference. Under this framework, positive trends were widespread, although only 37.57% of pixels crossed at least one fixed PRSEI class. This distinction indicates that trend direction and class transition capture different aspects of ecological change and should therefore be interpreted jointly.

Eco-regional divergence in ecological quality and trends

Eco-regional analysis further identified areas where positive trends coexist with persistent ecological constraints. The Hexi Desert–Gobi Region had the lowest 25-year mean PRSEI (0.225) and the slowest improvement rate (0.00270 yr^{-1}), while the Hexi Oasis–Corridor Agricultural Region retained the highest proportion of residual negative trends (2.07%). These patterns are consistent with persistent constraints related to water limitation, high evaporative demand, wind erosion, sandification, and intensive water use, indicating that positive trends do not necessarily imply the removal of ecological risks. By contrast, the Longnan–Tianshui Qinba Humid Mountain, Longzhong–Longdong Loess Hilly, and Gannan Alpine Mountain regions showed higher ecological baselines and/or faster improvement. Although these patterns are broadly consistent with more favourable hydrothermal conditions and previously reported restoration or soil-conservation processes, the available evidence does not permit attribution to specific policies or engineering interventions.

Management interpretation should jointly consider PRSEI baseline levels, class transitions, trend direction, and persistence tendencies. Low-value areas may still require continued protection despite positive trends, whereas small residual degradation patches are better suited to targeted monitoring and should not be interpreted as evidence of province-wide ecological decline.

Coupled spatial associations among water availability, soil, and land cover

The most robust OPGD finding was the consistent hierarchy of spatial associations across annual models. PRE, ST, and LULC remained the top three factors, with their principal pairwise interactions yielding q values of 0.889–0.907. This stable pattern supports a joint interpretation involving water availability, soil background, and land-cover configuration, while remaining insufficient to establish causal mechanisms.

A plausible ecological interpretation is that precipitation influences vegetation growth, surface moisture, and bare-ground exposure, while soil properties such as texture, water-holding capacity, salinity, and erodibility modulate responses to climatic inputs. LULC further differentiates major surface types, including cropland, grassland, forest, bare land, water bodies, and built-up areas^{41–46}. Although these pathways are consistent with previous studies, they should be regarded as mechanistic hypotheses because OPGD cannot resolve temporal mediation or intervention effects.

Interaction effects can be interpreted as context-dependent spatial associations: the relationship between PRE and PRSEI may vary across soil and LULC categories, and similar land-cover types may exhibit different PRSEI levels under contrasting moisture conditions. However, these interactions do not imply that changing a given factor will produce a specific PRSEI response. Soil-classification sensitivity further showed that ST q values remained similar after moderate aggregation but declined under coarser classifications. Thus, soil background is a robust contributor to spatial stratification, while the magnitude of its q value remains dependent on classification resolution.

Implications for ecological management across contrasting eco-regions

Management priorities should integrate PRSEI baseline conditions, class transitions, trend direction, and persistence tendencies. In the arid Hexi region, the results support continued protection of desert vegetation and oasis margins, targeted monitoring of residual degradation patches, and closer consideration of ecological water requirements in irrigation and groundwater management. Because this study did not quantify the effects of specific interventions, these measures should be viewed as evidence-based management priorities rather than prescriptive recommendations for particular engineering actions.

In the Loess Hilly, alpine, and humid mountain regions, relatively rapid ecological improvement suggests that restoration and conservation gains should be consolidated rather than assumed complete. Continued priorities include soil and water conservation, grassland and wetland protection, forest conservation, and monitoring of localised degradation. However, project-specific recommendations require independent information on restoration intensity, grazing pressure, water use, and protected-area management.

The enhanced joint associations among water availability, soil conditions, and LULC further suggest that ecological monitoring and intervention assessment should be stratified by environmental context. Such context-specific evaluation is better supported by the present evidence than uniform province-wide recommendations and provides a practical basis for prioritising future field validation and process-based research.

Limitations and future work

First, the 474.751 m grid is appropriate for long-term provincial-scale monitoring but may not resolve narrow valley oases, urban fringes, or small restoration patches. Higher-resolution Landsat, Sentinel, or Chinese satellite data should therefore be used for local validation while retaining the temporal continuity of the MODIS-based analysis.

Second, OPGD q values and interaction terms represent spatial associations rather than causal effects and cannot resolve temporal ordering, mediating pathways, or intervention responses. Future causal attribution should therefore integrate OPGD with process-based models, quasi-experimental designs, field observations, or matched restoration controls.

Third, the PM10-based TI is not dust-specific because PM10 includes both dust and non-dust sources. In addition, the archived TI grids for 2022–2024 extend beyond the period documented for LGHAP v2. Dust-specific interpretation therefore requires further validation using independent evidence such as AOD, dust-event frequency, or deposition records.

In addition, annual data on ecological engineering intensity, grazing pressure, groundwater abstraction, and irrigation water use were unavailable. Consequently, the low q value of NTL should not be interpreted as evidence that human activities exert little influence on ecological quality.

Finally, although unified PCA improves temporal comparability, it remains sensitive to global minimum–maximum extremes. The five fixed 0.2 classes should be interpreted as comparable mapping intervals rather than universal ecological thresholds, and CLCD provides external categorical consistency rather than field validation. Future work should therefore assess more robust scaling approaches, field-calibrated thresholds, and independent ecological datasets.

Conclusion

This study constructed a temporally comparable adaptive PRSEI for Gansu Province (2000–2024) using unified PCA and common scaling. The unified PC1 explained 79.12% of the total variance. Sensitivity analysis showed that long-term trend inference was highly dependent on PCA construction: pixel-level trend-direction agreement was only 39.73% between annual and unified PCA, but reached 99.90% between unified PCA and the

fixed-weight approach. These results highlight the importance of maintaining a consistent index framework in long-term ecological time-series analysis.

Under the unified-PCA framework, mean provincial PRSEI increased from 0.343 to 0.419 (slope = 0.00432 yr⁻¹; R² = 0.778). Positive trends covered 99.42% of the common-mask area, although only 37.57% of pixels increased by at least one fixed PRSEI class. Ecological improvement was therefore widespread but spatially heterogeneous: all five eco-regions showed positive trends, while the Hexi Desert–Gobi Region retained the lowest baseline and the Hexi Oasis–Corridor Agricultural Region the highest proportion of residual negative trends.

Index comparison showed that adaptive PRSEI remained highly consistent with original PRSEI spatially and showed greater consistency with mapped restoration transitions, but weaker consistency with degradation transitions than traditional RSEI. It should therefore be viewed as a complementary multi-stressor index for drylands rather than a universally superior replacement for RSEI. OPGD further identified PRE, ST, and LULC as the most stable spatial explanatory factors, although their *q* values indicate association rather than causality. Remaining uncertainties in TI interpretation, degradation detection, class calibration, and process attribution require further validation using independent atmospheric, field, and management data.

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Data availability The publicly available source datasets used in this study are listed in Table 1 and in the corresponding references. The derived annual PRSEI products, eco-regional statistics, trend and persistence classifications, and OPGD outputs generated during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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